



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2019

Mathematics Extension II

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Student Number at the top of each page
- Answer Section I- Multiple Choice on Answer Sheet provided
- Answer Section II – Free Response in a separate booklet for each question.
- NESA approved calculators and templates may be used.

Section I Multiple Choice

- 10 marks
- Attempt all questions
- Allow about 15 minutes for this section

Section II – Free Response

- 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.
- Allow about 2 hours 45 minutes for this section

Weighting: 30%

Section 1: Multiple Choice (10 marks)

Indicate your answer on answer sheet provided.

Allow approximately 15 minutes for this section.

Q1. A mass of 5 kg moves in a horizontal circle of radius 1.5 metres at a uniform angular speed of 4 radians per second. What is the centripetal force required for this motion?

- A. 40N
- B. 80N
- C. 120N
- D. 160N

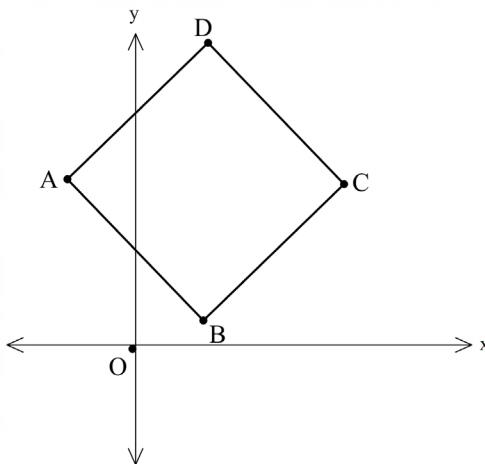
Q2. Find $\int \frac{dx}{x^2 - 6x + 13}$

- A. $\frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C$
- B. $\frac{1}{3} \tan^{-1} \left(\frac{x+3}{2} \right) + C$
- C. $\frac{1}{2} \tan^{-1} \left(\frac{x-2}{3} \right) + C$
- D. $\frac{1}{3} \tan^{-1} \left(\frac{x+2}{3} \right) + C$

Q3. What is the equation to the chord of contact to the ellipse $9x^2 + 16y^2 = 144$ from the point (8,6)?

- A. $3x + 4y = 6$
- B. $3x + 6y = 2$
- C. $9x + 16y = 144$
- D. $6x + 8y = 12$

- Q4. In the Argand diagram $ABCD$ is a square and the vertices A and B correspond to the complex numbers ω and z .



Which complex number corresponds to the diagonal BD ?

- A. $(\omega - z)(1 - i)$
 - B. $(\omega - z)(1 + i)$
 - C. $(z - \omega)(1 + i)$
 - D. $(\omega + z)(1 - i)$
- Q5. The polynomial $P(x) = x^3 + 2x^2 - 5x + 7$ has roots α , β and γ . Which polynomial has roots $\alpha + 1$, $\beta + 1$ and $\gamma + 1$?

- A. $x^3 - x^2 + 6x + 13 = 0$
- B. $x^3 - x^2 - 6x + 13 = 0$
- C. $x^3 - x^2 - 6x - 13 = 0$
- D. $x^3 + x^2 - 6x - 13 = 0$

Q6. What is an expression for the constant B such that $P(x) = (x - \alpha)^2 Q(x) + Ax + B$?

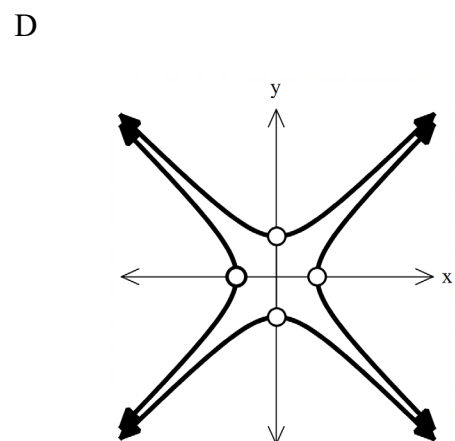
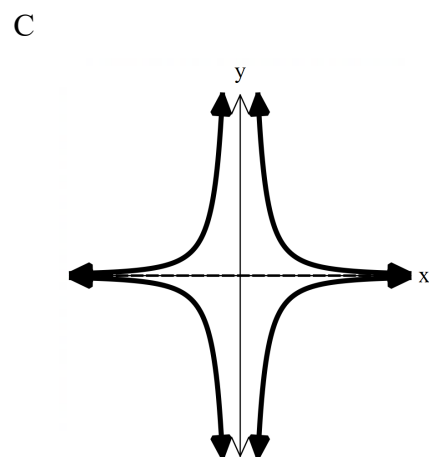
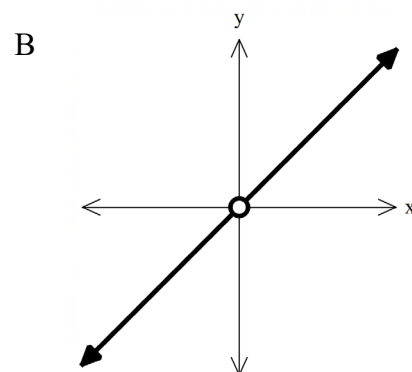
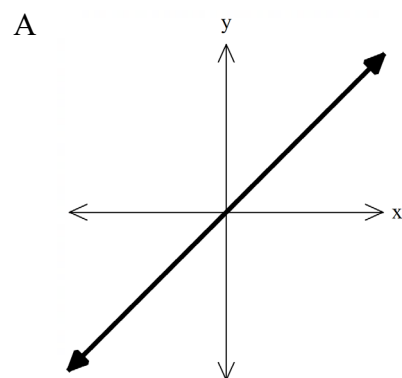
- A. $B = P(\alpha)$
- B. $B = P'(\alpha)$
- C. $B = P'(\alpha) - \alpha P(\alpha)$
- D. $B = P(\alpha) - \alpha P'(\alpha)$

Q7. Let ω be a complex cube root of -1. The value of $(1 + \omega - \omega^2)^3$ is:

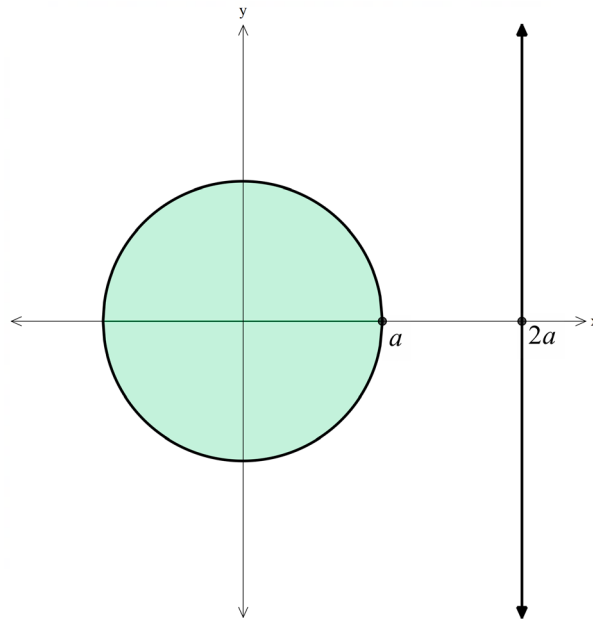
- A. 1
- B. -1
- C. 8
- D. -8

Q8. The equation $\frac{x}{y} + \frac{y}{x} = 2$ is an implicit function in x and y .

Which graph represents this implicit function?



Q9. The diagram below shows the circle $x^2 + y^2 = a^2$.



Solid A is formed by rotating the area enclosed by the circle around the line $x = 2a$.

The volume of solid A is V_A

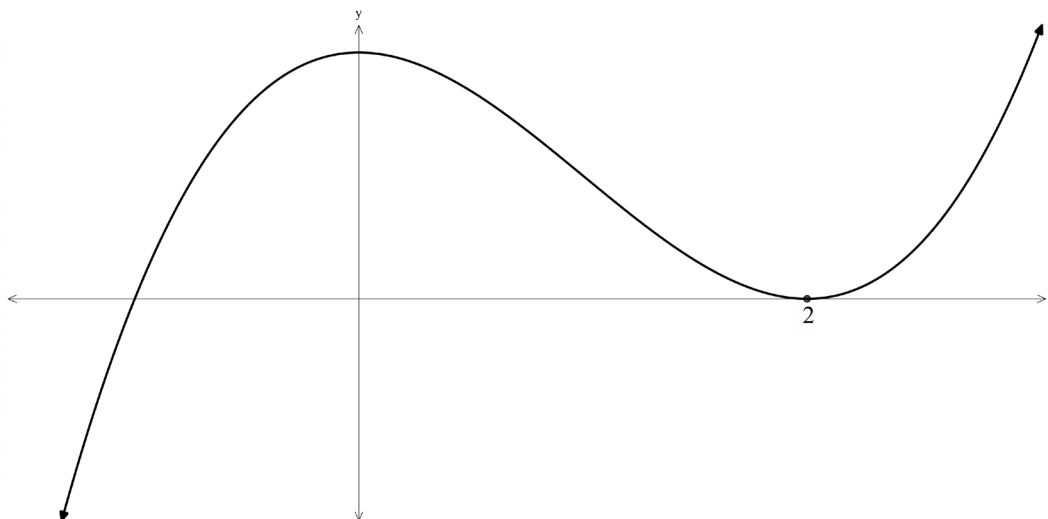
Another solid, Solid B , is formed by rotating the area enclosed by the circle around the line $x = 4a$.

The volume of solid B is V_B

Which of the following gives the correct volume of solid B ?

- A. $V_B = 2V_A$
- B. $V_B = 4V_A$
- C. $V_B = 8V_A$
- D. $V_B = 16V_A$

Q10. The graph below shows $y = f(u)$.



The function $g(x)$ is defined as $g(x) = \int_0^x f(u) \, du$.

Which of the statements below is true?

- A. $g(0) = 0$ and $g'(0) = 0$
- B. $g(0) > 0$ and $g'(2) = 0$
- C. $g''(0) = 0$ and $g'(2) = 0$
- D. $g'' > 0$ and $g'(2) = 0$

End of Multiple Choice

Section II Total Marks is 90

Attempt Questions 11 – 16.

Allow approximately 2 hours & 45 minutes for this section.

Answer all questions, starting each new question in a new booklet.

All necessary working must be shown in each and every question.

Question 11. – Start New Booklet

15 marks

- a. i Express $z = 2 - 2\sqrt{3}i$ in modulus-argument form 2
- ii Hence, otherwise, evaluate z^5 in simplest Cartesian form. 1
- b. Draw on an Argand diagram, the region defined by $\operatorname{Im}(2z + iz) \geq 2$ 2
[Your diagram must be at least one third of a page in size and must be **neat** and fully labelled.]
- c. Let $x = \alpha$ be a root of the polynomial $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$
where A and B are real numbers and $4A^2 \neq (2 + B)^2$
- i. Show that α cannot be 0, 1 or -1. 3
- ii. Show that $P\left(\frac{1}{\alpha}\right) = 0$ 1

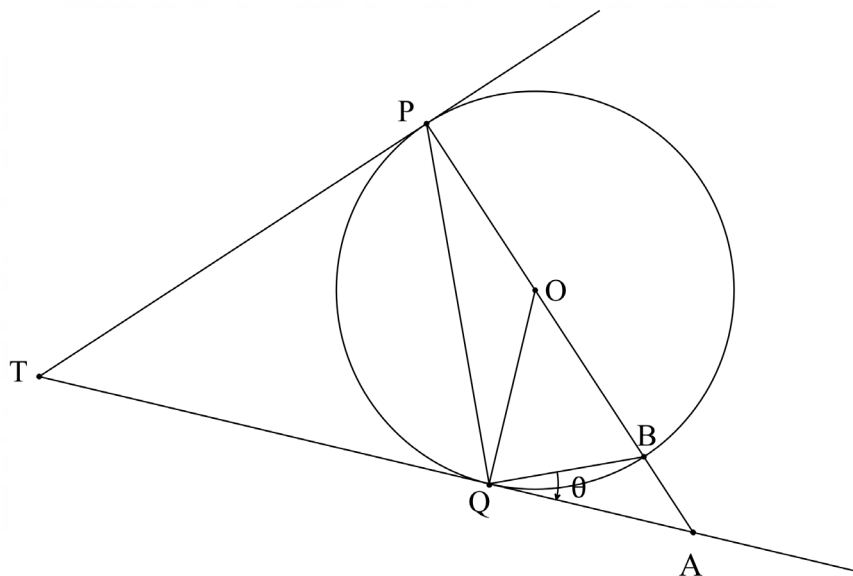
Question 11 continues on the next page.

Question 11 continued.

- d. From an external point T , tangents are drawn to a circle with centre O , touching the circle at P and Q . $\angle PTQ$ is acute.

The diameter PB produced meets the tangent TQ at A .

Let $\angle AQB = \theta$



(The diagram has been reproduced in your answer booklet. Answer this question on that page)

- | | | |
|------|--|----------|
| i) | Show that $\angle PTQ = 2\theta$ | 2 |
| ii) | Prove that $\triangle PBQ \parallel \triangle TOQ$ | 2 |
| iii) | Hence, show that $BQ \cdot OT = 2(OP)^2$ | 2 |

End of Question 11

Question 12. – Start New Booklet

15 marks

- a. A vehicle of mass 3000 kg is travelling around a horizontal circular road of radius 100m at a speed of 7.5 m/sec. Determine the centripetal force acting on the vehicle.

2

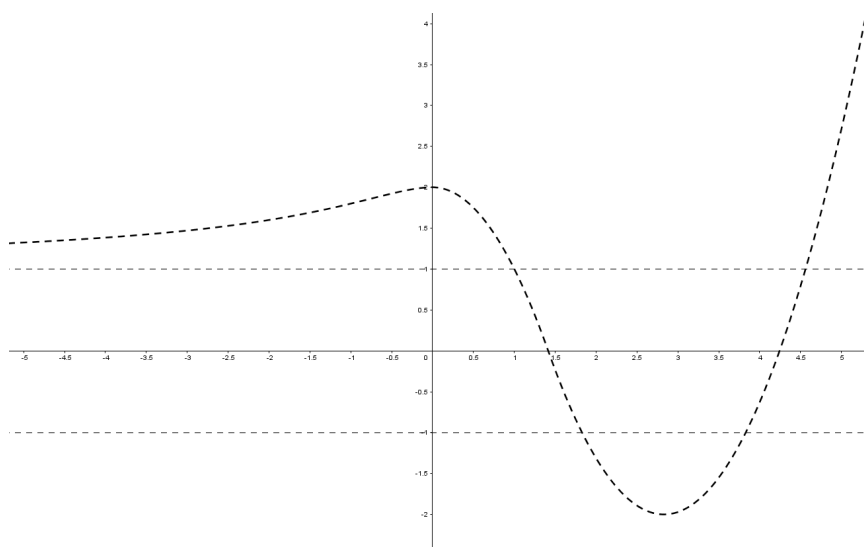
- b. i Write $\frac{2x^2 + 3x - 3}{x^2 - 1}$ in the form $A + \frac{B}{x - 1} + \frac{C}{x + 1}$.

2

- ii Hence find $\int \frac{2x^2 + 3x - 3}{x^2 - 1} dx$.

1

- c. The diagram shows the curve $f(x)$. The curve $f(x)$ is asymptotic to $y = 1$. The y -intercept is (0,2).



This curve $f(x)$ has been reproduced in your answer booklet.

Sketch the following curves showing all intercepts and asymptotes.

- i) $y = f(|x|)$ **1**
- ii) $y = \sqrt{f(x)}$ **2**
- iii) $y = \frac{1}{f(x)}$ **2**
- iv) $|y| = f(x)$ **2**
- v) $y = \ln [f(x)]$ **3**

End of Question 12

Question 13. – Start New Booklet

15 marks

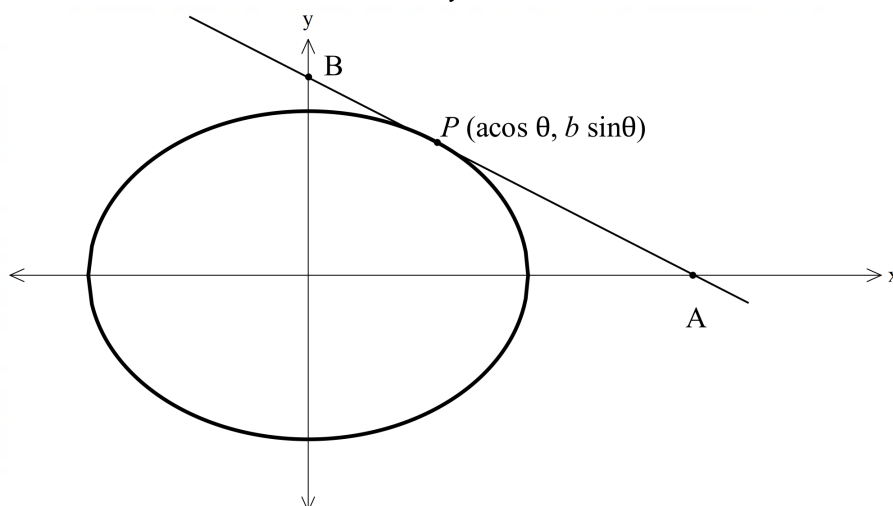
- a. Let $P(x) = x^4 + mx^3 + 36x^2 - 35x + n$ where m and n are real numbers.

It is given that $P(5) = 0$ and $P\left(\frac{1-\sqrt{3}i}{2}\right) = 0$.

- i) Show that $x^2 - x + 1$ is a factor of $P(x)$. 2
- ii) Find m and n . 2

- b. In the diagram below, $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The tangent at P cuts the x -axis at A and the y -axis at B .



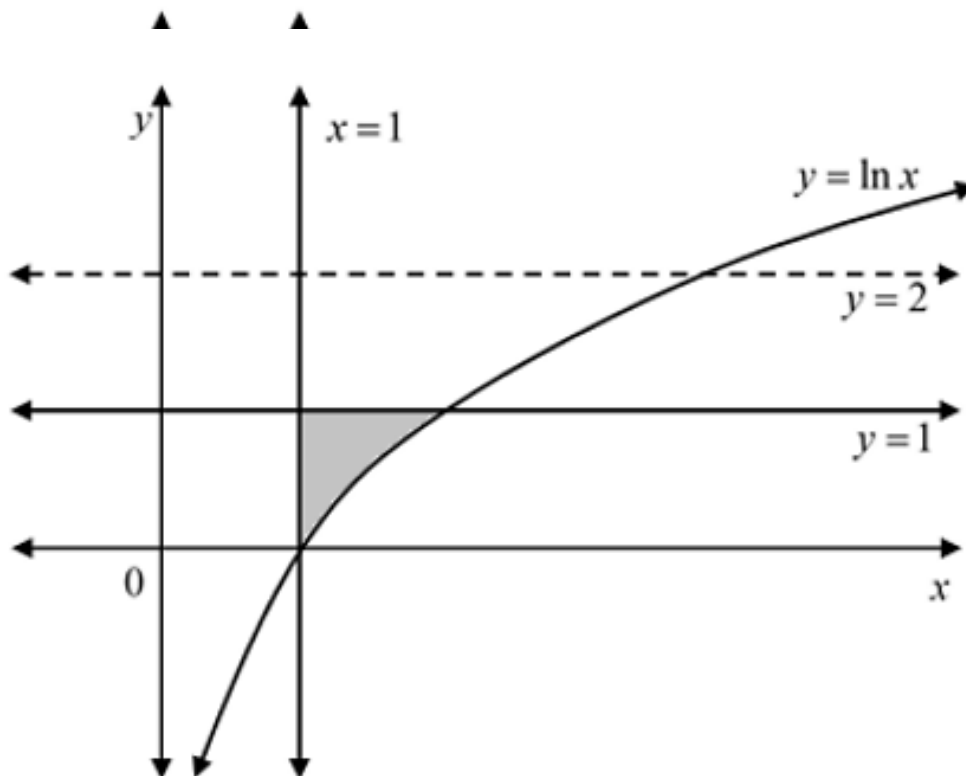
- i. Derive the parametric equation of the tangent at P in any form, and find the coordinates of A and B in parametric form. 2
 - ii. Show that $\frac{PA}{PB} = \tan^2 \theta$ 2
- c. A string is $0.5m$ long and will break if an object of mass exceeding $40kg$ is hung vertically from it. An object of mass $2kg$ is attached to one end of the string and it revolves around a horizontal circle with uniform speed. (Let gravity $g = 9.8 \text{ m/sec}^2$)
- i. Find the greatest angular velocity which may be imparted to the object without breaking the string 2
 - ii. Find the tangential speed at which this occurs. 1

Question 13 continues on the next page.

Question 13 continued.

Marks

- d. The region bounded by the curve $y = \ln x$, $x = 1$ and $y = 1$ is shaded in the diagram below. The region is rotated about the line $y = 2$ to form a solid.



Find the volume of the solid formed using the method of cylindrical shells.

4

End of Question 13

Question 14. – Start New Booklet

15 marks

a. Given $t = \tan x$,

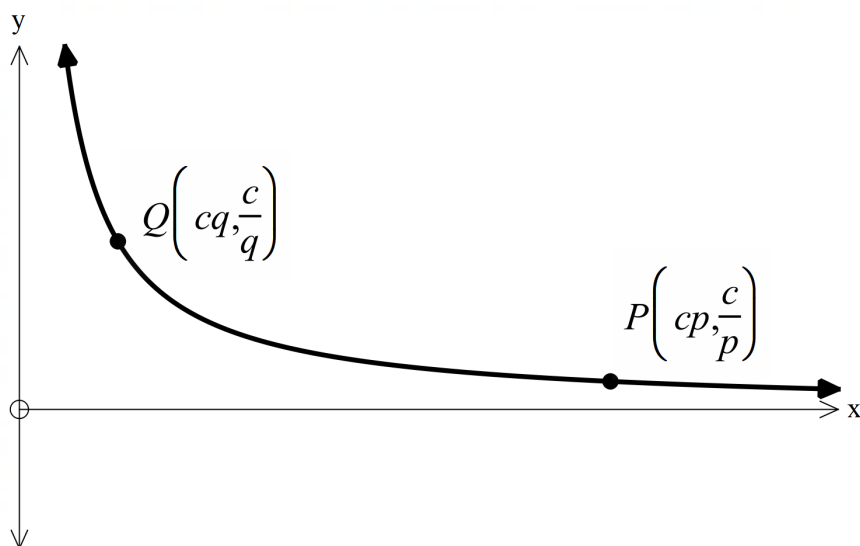
i. Show that $\frac{dx}{dt} = \frac{1}{1+t^2}$.

1

ii Use the substitution $t = \tan x$ to find $\int \frac{dx}{1 + \sin 2x}$

3

b. The variable points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $p > q > 0$ lie on the hyperbola $xy = c^2$. M is the midpoint of PQ .



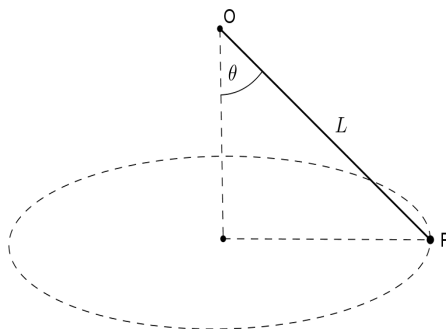
Given $p - q = 4$, find the equation of the locus of M .

3

Question 14 continues on next page.

Question 14 continued

- c. The diagram below shows a particle P of mass M kilograms suspended from a fixed point O by an inextensible string of length L metres.



P moves in a circle with centre directly below and distance h from O with uniform angular speed ω radians/sec.

The string makes an angle θ with the vertical and the acceleration due to gravity is $g \text{ ms}^{-2}$.

- i. Prove that the period of this motion is $2\pi\sqrt{\frac{h}{g}}$ 2
- ii. By considering the forces acting on the particle show that $\cos\theta = \frac{g}{L\omega^2}$. 3
- iii. The angular speed of the particle is increased to μ radians/sec.
At that speed the string makes an angle 2θ with the vertical.

Show that $\mu^2 = \frac{gL\omega^4}{2g^2 - L^2\omega^4}$. 3

End of Question 14

Question 15. – Start New Booklet

15 marks

a. Find

i $\int x e^{2x} dx$ **3**

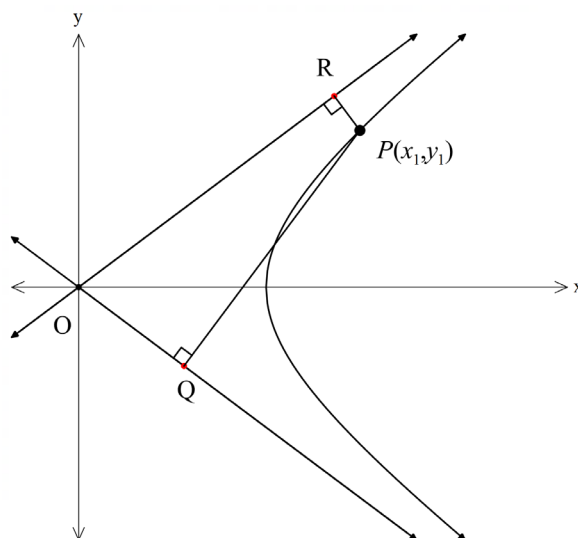
ii $\int_0^{\frac{\pi}{2}} \sin \theta (1 - \cos \theta)^2 d\theta$ **3**

b. i. Find the non-real solutions for of the equation $z^7 - 1 = 0$ **2**

ii Express $z^7 - 1$ as a product of linear and quadratic factors with real coefficients. **2**

iii Hence prove that $\cos \frac{\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{5\pi}{7} = \frac{1}{2}$ **2**

c. The diagram below shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



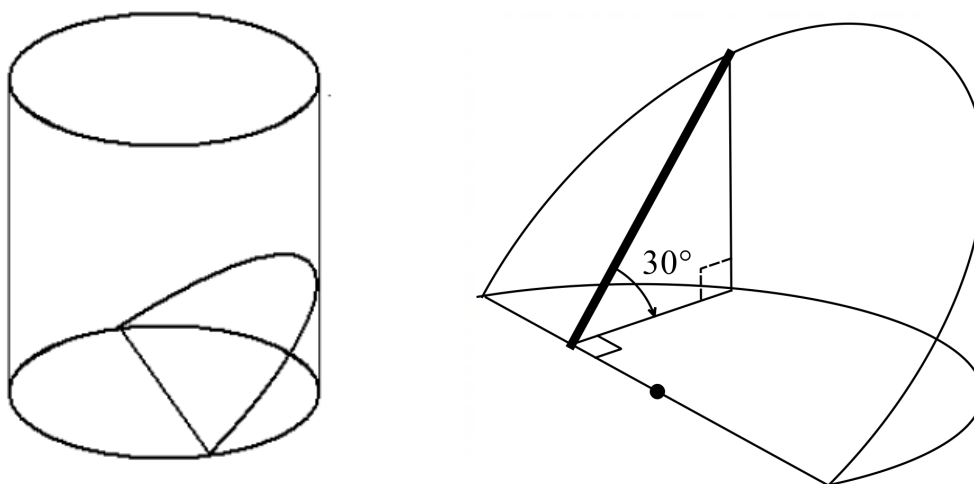
The point $P(x_1, y_1)$ is a point on the hyperbola.

The points Q and R lie on the asymptotes of the hyperbola such that $\angle PQO = \angle PRO = 90^\circ$. The eccentricity of the hyperbola is e .

Show that $PQ \times PR = \frac{b^2}{e^2}$.

3

End of Question 15



- a. A wedge is cut out of a circular cylinder of radius 5 cm. One plane is perpendicular to the axis of the cylinder. The other intersects the first plane at an angle of 30° along the diameter of the cylinder.

The cross section is a triangle with its base perpendicular to the diameter.

Find the volume of the shape.

4

$$\int \frac{\ln(1+x)}{1+x^2} dx = \int \ln(1+\tan\theta) d\theta$$

$$\text{let } x = \tan\theta \Rightarrow dx = \sec^2\theta$$

$$\therefore I = \int \frac{\ln(1+\tan x)}{1+\tan^2 x} \sec^2 \theta d\theta$$

$$= \int \frac{\ln(1+\tan x)}{\sec^2 \theta} \sec^2 \theta d\theta$$

- b. i Show

$$= \int \ln(1+\tan\theta) d\theta$$

1

- ii Hence, or otherwise, evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$

3

c. The integral I_n is defined as $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$ (for integers $n \geq 1$).

i. Show that $\int \frac{x}{(1+x^2)^n} dx = \frac{-1}{2(n-1)} \times \frac{1}{(1+x^2)^{n-1}} + C$ **2**

ii. By considering $\frac{1}{(1+x^2)^n} = \frac{1+x^2}{(1+x^2)^n} - \frac{x^2}{(1+x^2)^n}$, or otherwise,

show that $I_n = \frac{2n-3}{2(n-1)} I_{n-1} + \frac{1}{(n-1) \times 2^n}$ for $n \geq 2$. **3**

iii. Show that $I_n > \frac{1}{2^n}$ for $n \geq 1$. **2**

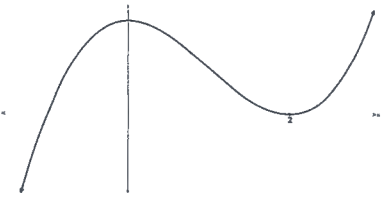
End of Examination

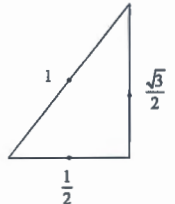
Q1. C Q2. A Q3.A Q4.A Q5.B Q6D* Q7C Q8B Q9A Q10.C

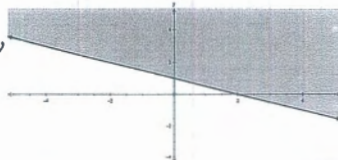
Q1	$F = m\omega^2$ $= 5 \times 1.5 \times 4^2$ $= 120N$	C
Q2	$\int \frac{dx}{x^2 - 6x + 13}$ $= \int \frac{dx}{x^2 - 6x + 9 + 4}$ $= \int \frac{dx}{(x-3)^2 + 2^2}$ $= \frac{1}{2} \tan^{-1} \frac{x-3}{2} + C$	A
Q3	$9x^2 + 16y^2 = 144$ $\frac{x^2}{16} + \frac{y^2}{9} = 1$ $\frac{x x_0}{16} + \frac{y y_0}{9} = 1$ $\frac{8x}{16} + \frac{6y}{9} = 1$ $\frac{x}{2} + \frac{2y}{3} = 1$ $3x + 4y = 6$	A
Q4	$\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$ $\overrightarrow{BA} = \omega - z$ $\overrightarrow{AB} = z - \omega$ $\overrightarrow{AD} = i\overrightarrow{AB} = i(z - \omega)$ $\overrightarrow{BD} = (\omega - z) + i(z - \omega)$ $= (\omega - z)(i - i)$	A

Q5	$X = x + 1 \Rightarrow X - 1$ $(X - 1)^3 + 2(X - 1)^2 - 5(X - 1) + 7$ $= X^3 - 3X^2 + 3X - 1 + 2X^2 - 4X + 2$ $- 5X + 5 + 7$ $= X^3 - X^2 - 6X + 13$ $= x^3 - x^2 - 6x + 13$	B
Q6	$P(x) = (x - \alpha)^2 Q(x) + Ax + B$ let $x = \alpha$ $P(\alpha) = A\alpha + B$ $P'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x) + A$ $P'(\alpha) = A$ $P(\alpha) - \alpha P'(\alpha) = A\alpha + B - \alpha A$ $B = P(\alpha) - \alpha P'(\alpha)$	D
Q7	$z^3 + 1 = 0$ $(z + 1)(1 - z + z^2) = 0$ if ω is a complex root $\therefore 1 - \omega + \omega^2 = 0$ $1 - \omega = -\omega^2$ $(1 + \omega - \omega^2)^3$ $= (1 + \omega + 1 - \omega)^3$ $= (2)^3$ $= 8$	C
Q8	$\frac{x}{y} + \frac{y}{x} = 2$ $x^2 + y^2 = 2xy$ $x^2 - 2xy + y^2 = 0$ $(x - y)^2 = 0$ $y = x$ $x \neq 0; y \neq 0$	B

Q9	Short solution Taking slices of width Δy perpendicular to axis of rotation: $V_A = \pi \int_{-a}^a \left[(2a+x)^2 - (2a-x)^2 \right] dy$ $= \pi \int_{-a}^a (8ax) dy$ $= 4\pi^2 a^3 \text{ units}^3 \text{ (given)}$ $V_B = \pi \int_{-a}^a \left[(4a+x)^2 - (4a-x)^2 \right] dy$ $= \pi \int_{-a}^a (16ax) dy$ $= 2 \times V_A$ $= 8\pi^2 a^3 \text{ units}^3$ Long solution Volume = $\pi (R^2 - r^2)$ height $= \pi (R+r)(R-r) \delta y$ $R = 4a + x$ $r = 4a - x$ $V = \pi 8a \times 2x \delta y$ $= 16\pi a \int_{-a}^a x dy$ $= 2 \times 16\pi a \int_0^a \sqrt{a^2 - y^2} dy$ $= 32\pi a \int_0^a \sqrt{a^2 - y^2} dy$ let $y = a \sin \theta$ $dy = a \cos \theta$ $y = a \Rightarrow a \sin \theta = a \Rightarrow \theta = \frac{\pi}{2}$ $= 32\pi a \int_0^{\frac{\pi}{2}} a \cos \theta a \cos \theta d\theta$ $= 32\pi a^3 \int_0^{\frac{\pi}{2}} \frac{1}{2} (\cos 2\theta + 1) d\theta$ $= 16\pi a^3 \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\frac{\pi}{2}}$ $= 16\pi a^3 \left\{ \left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - (0 + 0) \right\}$ $= 8\pi^2 a^3$	A
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Q10	 $g''(0) = 0$ and $g'(2) = 0$ Given $g(x) = \int_0^x f(x) dx$ then $f(x) = g'(x)$ $\therefore f'(x) = g''(x)$ From graph $f'(0) = 0 \therefore g''(0) = 0$ Similarly as $f(x) = g'(x)$ $\therefore f(2) = g'(2) = 0$ Option C	C
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a-i	m and n are real therefore roots are in conjugate pairs. Roots are $5, \left(\frac{1-\sqrt{3}i}{2}\right), \left(\frac{1+\sqrt{3}i}{2}\right)$ Factors $(x-5)\left(x-\frac{1-\sqrt{3}i}{2}\right), \left(x-\frac{(1)+\sqrt{3}i}{2}\right)$ $(x-z)(x-\bar{z}) = x^2 - 2\operatorname{Re}(z) + (z)^2$ $\left(x-\frac{1-\sqrt{3}i}{2}\right)\left(x-\frac{(1)+\sqrt{3}i}{2}\right)$ $= \left(x^2 - 2 \times \frac{1}{2} \times x + 1\right)$ 	2 marks – correct solution
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ai	$ z = \sqrt{2^2 + (-2\sqrt{3})^2}$ $= \sqrt{16}$ $= 4$ $\operatorname{Arg}(z) = -\tan^{-1}\left(\frac{2\sqrt{3}}{2}\right)$ $= -\tan^{-1}(\sqrt{3})$ $= -\frac{\pi}{3}$ $\therefore z = 4\operatorname{cis}\left(-\frac{\pi}{3}\right)$	2 marks-correct solution 1 mark-correct mod or arg
aii	$z^5 = \left[4\operatorname{cis}\left(-\frac{\pi}{3}\right)\right]^5$ $= 4^5 \operatorname{cis}\left(-\frac{5\pi}{3}\right)$ $= 1024 \left(\cos\left(-\frac{5\pi}{3}\right) + i\sin\left(-\frac{5\pi}{3}\right)\right)$ $= 1024 \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ $= 1024 \left(\frac{1}{2} + i\sqrt{\frac{3}}{2}\right)$ $= 512 + 512\sqrt{3}i$	1 mark-correct answer
b	let $z = x + iy$ $2z + iz = 2x + 2iy + iy + i^2y$ $= 2x - y + i(x + 2y)$ $\operatorname{Im}(2z + iz) = x + 2y$ $\therefore x + 2y \geq 2$ 	2 marks-correct solution 1 mark-correct inequality but incorrect region
ci	$P(0) = 1 \therefore \alpha \neq 0$ $P(1) = 2 + 2A + B$ if $\alpha = 1$ then $P(1) = 0$ then $2 + B = -2A$ $(2 + B)^2 = (-2A)^2$ $(2 + B)^2 = 4A^2$ but $(2 + B)^2 \neq 4A^2$ $\therefore \alpha \neq 1$ $P(-1) = 2 - 2A + B$ if $\alpha = -1$ then $P(-1) = 0$ then $2 + B = 2A$ $(2 + B)^2 = (2A)^2$ $(2 + B)^2 = 4A^2$ but $(2 + B)^2 \neq 4A^2$ $\therefore \alpha \neq -1$	3 marks –correct solution 2 marks-ONLY one error in correct progress to proof 1 mark- ONLY one correct root shown

cii	$P\left(\frac{1}{\alpha}\right) = \left(\frac{1}{\alpha}\right)^4 + A\left(\frac{1}{\alpha}\right)^3 + B\left(\frac{1}{\alpha}\right)^2 + A\left(\frac{1}{\alpha}\right) + 1$ $= \frac{1}{\alpha^4} + \frac{A}{\alpha^3} + \frac{B}{\alpha^2} + \frac{A}{\alpha} + 1$ $= \frac{1}{\alpha^4}(A\alpha + B\alpha^2 + A\alpha^3 + \alpha^4)$ and $P(\alpha) = 0$ ie $A\alpha + B\alpha^2 + A\alpha^3 + \alpha^4 = 0$ $\therefore P\left(\frac{1}{\alpha}\right) = \frac{1}{\alpha^4} \times 0$ $= 0$	1 mark correct solution
di	$\angle BPQ = \theta$ (alternate segment) $\angle BOQ = 2\theta$ ($\angle$ centre is $2 \times \angle$ at circumference) $\angle POQ + \angle BOQ = 180^\circ$ (supplementary) $\angle TPO = \angle TQO = 90^\circ$ (tangent $\perp$ radii) $\angle PTQ + \angle POQ = 180^\circ$ ($\angle$ sum Δ) $\therefore \angle PTQ = \angle BOQ = 2\theta$	2 marks-correct solution 1 mark- one correctly used relevant theorem in a progress to proof.
dii	$\Delta PBQ \Delta TOQ$ $\angle PQB = 90^\circ$ ($\angle$ in semicircle) $\angle OQT = 90^\circ$ (tangent $\perp$ radii) $\therefore \angle PQB = \angle OQT$ ΔPOQ isosceles (equal radii) $\angle TOQ = \frac{180 - 2\theta}{2}$ (base $\angle$ of Isos Δ) $= 90 - \theta$ $\angle PBQ = 90 - \theta$ ($\angle$ sum Δ) $\therefore \angle TOQ = \angle PBQ$ $\therefore \Delta PBQ \parallel \Delta TOQ$ (equiangular)	2 marks-correct solution 1 mark- one correctly used relevant theorem in a progress to proof.
diii	$\frac{OT}{BP} = \frac{OQ}{BQ}$ (corresponding sides in equal radii) $BP = OP + OB$ (diameter-given) $OP = OB = OQ$ (equal radii) $\therefore BP = 2OP$ hence $\frac{OT}{2OP} = \frac{OQ}{BQ}$ $BQ \cdot OT = 2OP \cdot OP$ $= 2(OP)^2$	2 marks-correct solution 1 mark- one correctly used relevant theorem in a progress to proof.

Question 12

Marks

$$(a) F_c = \frac{mv^2}{r} \Rightarrow F_c = \frac{3000 \times 7.5^2}{100}$$

2 Marks

$$= 1687.5 \text{ N}$$

Correct solution

1 Mark

Correct substitution

$$(b) (i) \frac{2x^2 + 3x - 3}{x^2 - 1} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x-1}$$

2 Marks

Correct solution

$$2x^2 + 3x - 3 = A(x^2 - 1) + B(x+1) + C(x-1)$$

$$x=1 \Rightarrow B=1; x=-1 \Rightarrow C=2; x=0 \Rightarrow A=2$$

1 Mark

Obtains one

correct value

for A or B or C

$$\frac{2x^2 + 3x - 3}{x^2 - 1} = 2 + \frac{1}{x-1} + \frac{2}{x+1}$$

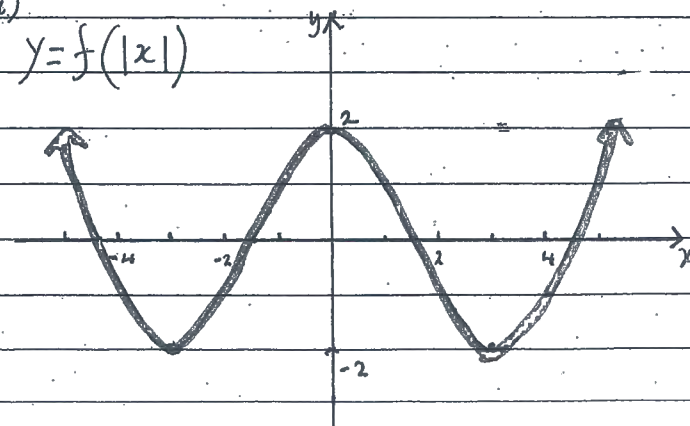
$$(ii) \int \frac{2x^2 + 3x - 3}{x^2 - 1} dx = \int \left\{ 2 + \frac{1}{x-1} + \frac{2}{x+1} \right\} dx$$

1 Mark

Correct solution

$$= 2x + \log_e(x-1) + 2\log_e(x+1)$$

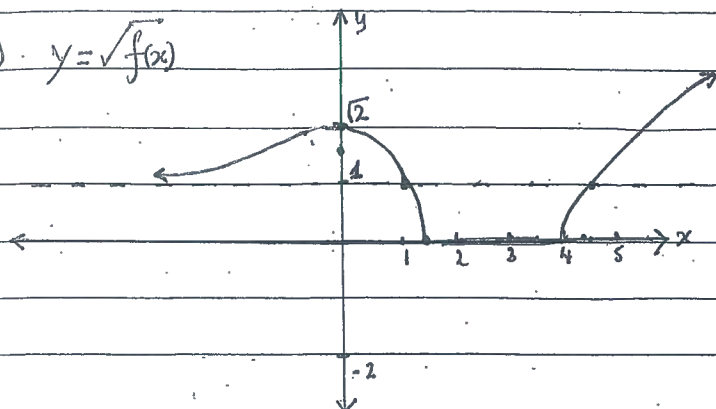
(c) (i)



1 Mark

Correct graph

(ii) $y = \sqrt{f(x)}$



Marks

2 Marks

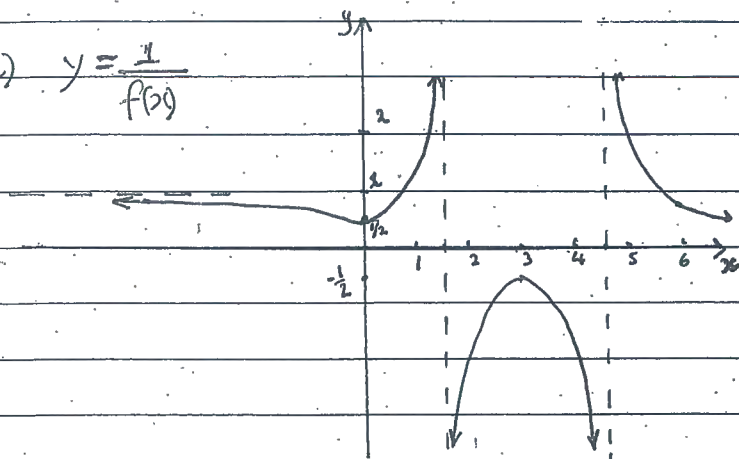
Correct graph

1 Mark

Correct LHS

including y
intercept
explicit or
implicit.

(iii) $y = \frac{1}{f(x)}$



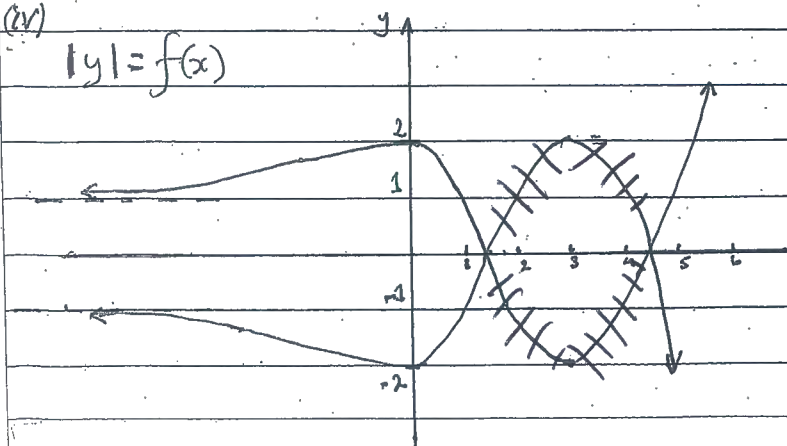
2 Marks

Correct graph

1 Mark

Two of three
features
correct

(iv) $|y| = f(x)$



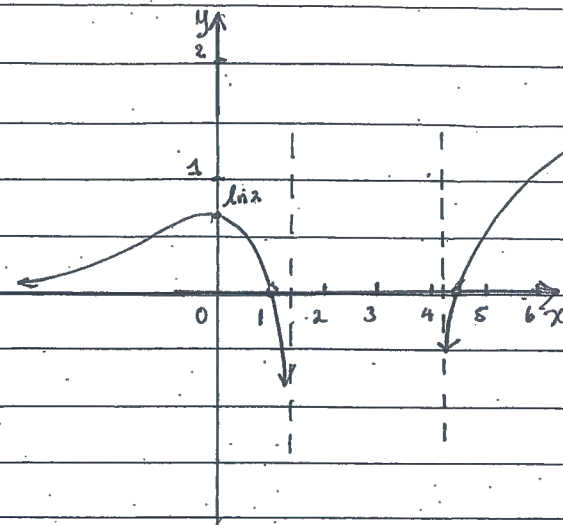
2 Marks

Correct graph

1 Mark

Correct LHS and
RHS but excluded
graph for 1.55x4.5

(v)



Marks

3 Marks

Correct graph

2 Marks

Graph with two
of three
asymptotes

correct And

correct y

intercept,

explicit or

implicit

1 Mark

Two correct

components

other than

asymptotic

behaviour of

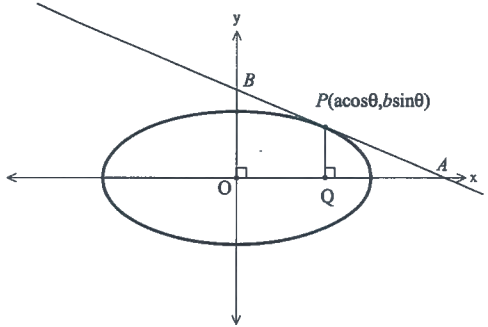
graph

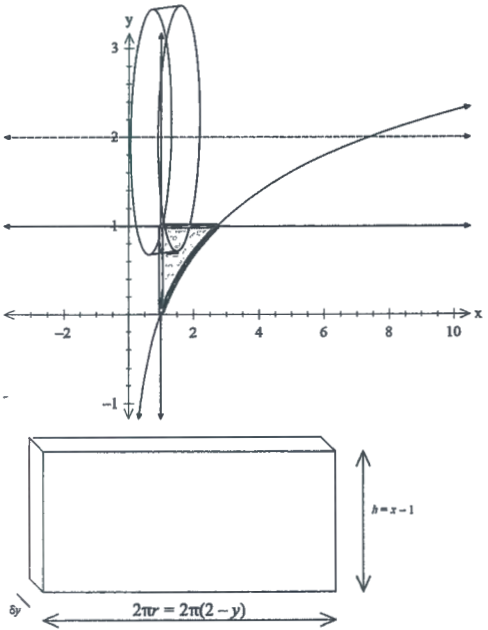
	$P(x) = (x-5)(x^2 - x + 1)(x+a)$ $x = 0$ $n = -5a$ $x = 1$ $1 + m + 36 - 35 + n = -4 - 4a$ $m + n + 2 = -4 - 4a$ $m - 5a = -6 - 4a$ $m - a = -6$ $m = a - 6$ $x = -1$ $1 - m + 36 + 35 + n = -6 \times 3 \times (a-1)$ $n - m + 72 = 18 - 18a$ $-5a - m + 18a = -54$ $13a - (a-6) = -54$ $12a + 6 = -54$ $12a = -60$ $a = -5$ $\therefore \quad m = -11 \quad n = 25$	
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Question 13

	$P(x) = x^4 + mx^3 + 36x^2 - 35x + n$ $P(x) = (x-5)(x-z)(-\bar{z})(x^2 + ax + b)$ $(x-z)(x-\bar{z}) = x^2 - 2\operatorname{Re}(z) + (z)^2$ $z = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ $ z = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$ $p(x) = (x-5)\left(x^2 - 2 \times \frac{1}{2}x + (1)^2\right)(ax+b)$ $P(x) = (x-5)(x^2 - x + 1)(x+b)$ Method 1. Substitution. Let $x = 1$ $LHS = 1 \times -4 \times (1 - \beta) = 4\beta - 4$ $RHS = 1 + m + 36 - 35 + n$ $\therefore 4\beta - 6 = m + n \text{ (1)}$ a Let $x = 0$ $LHS = 5\beta$ $RHS = n$ $\therefore 5\beta = n \text{ (2)}$ Let $x = 1$ $LHS = 3 \times -6 \times (-1 - \beta)$ $= 18 + 18\beta$ $RHS = 1 - m + 36 + 35 + n$ $= 72 + n - m$ $\therefore 18\beta - 54 = n - m \text{ (3)}$ (1) + (3) $22\beta - 60 = 2n$ Subst (2) $22b^2 - 60 = 10\beta$ $12\beta = 60$ $\beta = 5$ $\therefore n = 5\beta = 25$ $4 \times 5 - 6 = 25 + m \text{ from (1)}$ $m = -11$	i) 2 marks ii) 2 marks 1 mark - Subst method - removal of one term in simul eqn 1 mark - correct expansion
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	Could also be done by equating coefficients $(x-5)(x^2-x+1)(x-\beta) = x^4 - (6+\beta)x^3 + 6(\beta+1)x^2 - (5+6\beta)x + 5\beta$ $\therefore n = 5\beta$ Coefficient of x^2 $6(\beta+1) = 36$ $\beta = 5$ $\therefore n = 25$ $m = -(6+\beta) = -11$	
	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{b^2x}{a^2y}$ $y - y_1 = -\frac{b^2x_1}{a^2y_1}(x - x_1)$ $a^2y_1y - a^2y_1^2 = -b^2x_1x + b^2x_1^2$ $b^2x_1x + a^2y_1y = a^2y_1^2 + b^2x_1^2$ $\div a^2\beta$ $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$ $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = 1$ $x = a\cos\theta \quad y = b\sin\theta$ $\frac{a\cos\theta x}{a^2} + \frac{b\sin\theta y}{b^2} = 1$ $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$ at $x = 0 \quad y = \frac{b}{\sin\theta}$ at $y = 0 \quad x = \frac{a}{\cos\theta}$	2 marks 1 mark

	 Ratio of transversals on parallel lines $\frac{AP}{AQ} = \frac{PB}{QO}$ $\therefore \frac{AP}{PB} = \frac{AQ}{QO}$ At $y = 0$ (from equation to tangent) $x = \frac{a}{\cos\theta}$ $\therefore AQ = \frac{a}{\cos\theta} - a\cos\theta = \frac{a(1 - \cos^2\theta)}{\cos\theta}$ $QO = a\cos\theta$ $\frac{AP}{QO} = \frac{a(1 - \cos^2\theta)}{\cos\theta} \times \frac{1}{a\cos\theta} = \frac{\sin^2\theta}{\cos^2\theta}$ $= \tan^2\theta$ $\frac{PA}{PB} = \tan^2\theta$	2 marks 1 mark
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Q.3 c-1	String will break above $T = mg = 40 \times 9.8 = 392N$ $F = mr\omega^2$ $392 = \frac{1}{2} \times 2 \times \omega^2$ $\omega = \sqrt{392} = 14\sqrt{2} \text{ m/s} \approx 19.8$	2 marks
c-1	$v = r\omega$ $v = \frac{1}{2} \times 14\sqrt{2}$ $= 7\sqrt{2} \text{ m/s} = 9.9$	1 mark
13-d		4 marks 3 marks 2 marks 1 mark - Correct δV

	$\delta V = 2\pi r \cdot h \cdot w$ $= 2\pi(2-y)(x-1)\delta y$ $y = \ln x \Rightarrow x = e^y$ $\delta V = 2\pi(2-y)(e^y - 1)\delta y$ $V \cong \lim_{\delta y \rightarrow 0} \sum_0^1 2\pi(2-y)(e^y - 1)\delta y$ $V = 2\pi \int_0^1 (2-y)(e^y - 1)dy$ $= 2\pi \int_0^1 (2e^y - 2 - ye^y + y)dy$ $= 2\pi \left\{ \left[2e^y - 2y + \frac{y^2}{2} \right]_0^1 - \int_0^1 ye^y dy \right\}$ Let $m = ye^y$ $\frac{dm}{dy} = e^y + ye^y$ $\therefore ye^y = \int e^y + ye^y dy$ $\int ye^y dy = ye^y - \int e^y dy = ye^y - e^y + C$ $V = 2\pi \left[2e^y - 2y + \frac{y^2}{2} - ye^y + e^y \right]_0^1 = 2\pi$ $= 2\pi \left\{ \left(3e - 2 + \frac{1}{2} - e \right) - (3 - 0 - 0 - 0) \right\}$ $= 2\pi \left(2e - \frac{9}{2} \right)$ $= (4e - 9)\pi u^3$	
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ai	$\frac{dt}{dx} = \sec^2 x$ $= 1 + \tan^2 x$ $= 1 + t^2$ $\therefore \frac{dx}{dt} = \frac{1}{1+t^2}$	1 mark correct solution
aii	$\int \left(\frac{1}{1 + \left[\frac{2t}{1+t^2} \right]} \times \left(\frac{dt}{1+t^2} \right) \right)$ $= \int \frac{1+t^2}{1+t^2+2t} \times \frac{dt}{1+t^2}$ $= \int \frac{dt}{(1+t)^2}$ $= \int (1+t)^{-2} dt$ $= -(1+t)^{-1} + c$ $= \frac{1}{1+\tan x} + c$	3 marks-correct solution 2 marks- correct substitutions and integration with ONLY one error in correct progress to answer. 1 mark-correct substitutions
b	$M_{pe} = \left(\frac{c(p+q)}{2}, \frac{c}{2} \left(\frac{1}{p} + \frac{1}{q} \right) \right)$ $= \left(\frac{c(p+q)}{2}, \frac{c}{2} \frac{(p+q)}{pq} \right)$ $p-q=4 \Rightarrow p=4+q$ $x = \frac{c}{2}(4+2q)$ $x = c(2+q)$ $\frac{2x}{c} = 2p-2$ $\frac{x}{c} = 2 = q$ $q = \frac{x}{c} - 2$ $y = \left(\frac{1}{2} \left(\frac{cp+cp}{pq} \right) \right)$ $= \frac{c}{2} \left(\frac{p+q}{pq} \right)$ $= \frac{c}{2} \left(\frac{4+2q}{(4+q)q} \right)$ $= \frac{2c}{2} \left(\frac{2+q}{(4+q)q} \right)$ $= \frac{c \left(\frac{x}{c} \right)}{\left(\frac{x}{c} + 2 \right) \left(\frac{x}{c} - 2 \right)}$ $= \frac{x}{\left(\frac{x}{c} \right)^2 - 4}$ $= \frac{x}{\left(\frac{x^2 - 4c^2}{c^2} \right)}$ $\therefore \text{Locus} \Rightarrow y = \frac{xc^2}{x^2 - 4c^2}$	3 marks- correct solution 2 marks- one error in correct progress to locus using given info without QS 1 mark-one correct use of given info simplify x or y

ci	$r = h \tan \theta$ $T \sin \theta = M(h \tan \theta) \omega^2$ $T \cos \theta = Mg$ $\therefore \tan \theta = \frac{h \tan \theta \omega^2}{g}$ $\omega^2 = \frac{g}{h}$ $\omega = \sqrt{\frac{g}{h}}$	$period = \frac{2\pi}{\omega}$ $= \frac{2\pi}{\sqrt{\frac{g}{h}}}$ $= 2\pi \sqrt{\frac{h}{g}}$	2 marks-correct solution 1 mark-correct equations for motion
cii	$T \cos \theta = Mg$ $\cos \theta = \frac{Mg}{T}$ $T \sin \theta = Mr \omega^2$ $T = \frac{mr \omega^2}{\sin \theta}$	$\cos \theta = \frac{Mg}{mr \omega^2}$ $= \frac{\sin \theta}{\frac{Mg \sin \theta}{mr \omega^2}}$ $= \frac{g \sin \theta}{r \omega^2}$ $= \frac{g \left(\frac{r}{L} \right)}{r \omega^2} \text{ since } \sin \theta = \frac{r}{L}$ $= \frac{g}{L \omega^2}$	3 marks- correct solution 2 marks- correct use of equations of motion to create equation in cos and significant relevant progress to expression 1 mark-demonstration of use of equations of motion to obtain required result
ciii	$\cos 2\theta = 2\cos^2 \theta - 1$ $\cos 2\theta = \frac{g}{L \mu^2}$ $\therefore \frac{g}{L \mu^2} = 2 \left(\frac{g}{L \omega^2} \right)^2 - 1 \text{ from part ii}$ $= \frac{2g^2}{L^2 \omega^4} - 1$ $= \frac{2g^2 L^2 \omega^4}{L^2 \omega^4}$ $\therefore \mu^2 = \frac{gL^2 \omega^4}{L(2g^2 - L^2 \omega^4)}$ $= \frac{gL \omega^4}{2g^2 - L^2 \omega^4}$		3 marks- correct solution 2 marks- correct use of both cos equations and significant relevant progress to expression 1 mark-recognise the of use of both cos result to create/equate

Q15

a-1)	$\int x e^{2x} dx$ $u = x \quad v' = e^{2x}$ $u' = 1 \quad v = \frac{1}{2} e^{2x}$ $I = \frac{x}{2} e^{2x} - \frac{1}{2} \int e^{2x} dx$ $= \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C$	3 marks – correct solution 2 marks – one error 1 mark – correct separation into parts
a-11)	$\int_0^{\frac{\pi}{2}} \sin \theta (1 - \cos \theta)^2 d\theta$ $= \int_0^{\frac{\pi}{2}} \sin \theta (1 - 2\cos \theta + \cos^2 \theta) d\theta$ $= \int_0^{\frac{\pi}{2}} \sin \theta - 2\sin \theta \cos \theta + \sin \theta \cos^2 \theta d\theta$ $= \int_0^{\frac{\pi}{2}} \sin \theta - \sin 2\theta + \sin \theta \cos^2 \theta d\theta$ $= \left[-\cos \theta + \frac{\cos 2\theta}{2} - \frac{\cos^3 \theta}{3} \right]_0^{\frac{\pi}{2}}$ $\left[-\cos \frac{\pi}{2} + \frac{\cos \pi}{2} - \frac{\cos^3 \frac{\pi}{2}}{3} \right] - \left[-\cos 0 + \frac{\cos 0}{2} - \frac{\cos^3 0}{3} \right]$ $= \left[0 - \frac{1}{2} + 0 \right] - \left[-1 + \frac{1}{2} - \frac{1}{3} \right]$ $= -\frac{1}{2} + \frac{1}{2} + \frac{1}{3} = \frac{1}{3}$ Better version	3 marks - Correct solution 2 marks - Correct use of trig identity 1 mark - One correct integration

	let $u = 1 - \cos \theta$ $du = \sin \theta d\theta$ $\theta = \frac{\pi}{2} \Rightarrow u = 1$ $\theta = 0 \Rightarrow u = 0$ $\int_0^1 u^2 du$ $= \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{3}$	
		2 marks – all solutions correct 1 mark –
	$P(z) = (z - 1)(z - \omega_1)(z - \omega_2)(z - \omega_3)(z - \omega_4)(z - \omega_5)(z - \omega_6)$ $P(z) = (z - 1)(z - \omega_1)(z - \omega_6)(z - \omega_2)(z - \omega_5)(z - \omega_3)(z - \omega_4)$ $= (z - 1)(z - \omega_1)\left(z - \overline{\omega_1}\right)(z - \omega_2)\left(z - \overline{\omega_2}\right)(z - \omega_3)\left(z - \overline{\omega_3}\right)$ $= (z - 1)\left(z^2 - 2\operatorname{Re}(\omega_1)z + \omega_1 ^2\right)\left(z^2 - 2\operatorname{Re}(\omega_2)z + \omega_2 ^2\right)\left(z^2 - 2\operatorname{Re}(\omega_3)z + \omega_3 ^2\right)$ $= (z - 1)\left(z^2 - 2\cos \frac{2\pi}{7} z + 1\right)\left(z^2 - 2\cos \frac{4\pi}{7} z + 1\right)\left(z^2 - 2\cos \frac{6\pi}{7} z + 1\right)$	
	2marks	
	1 mark	

$\Sigma \alpha = -\frac{b}{a} = \frac{0}{1}$ $1 + \operatorname{cis} \frac{2\pi}{7} + \operatorname{cis} \frac{4\pi}{7} + \operatorname{cis} \frac{6\pi}{7} + \operatorname{cis} \frac{8\pi}{7} + \operatorname{cis} \frac{10\pi}{7} + \operatorname{cis} \frac{12\pi}{7} = 0$ $\therefore$ $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{10\pi}{7} + \cos \frac{12\pi}{7} = -1$ $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos -\frac{6\pi}{7} + \cos -\frac{4\pi}{7} + \cos -\frac{2\pi}{7} = -1$ as $\cos x = \cos(-x)$ $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{6\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} = -1$ $2\left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}\right) = -1$ $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$ as $\cos(\pi - x) = -\cos x$ $-\cos \frac{5\pi}{7} - \cos \frac{3\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$ $\cos \frac{5\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{\pi}{7} = \frac{1}{2}$	
2 marks 1 mark	

3 marks - correct solution
1 - mark -

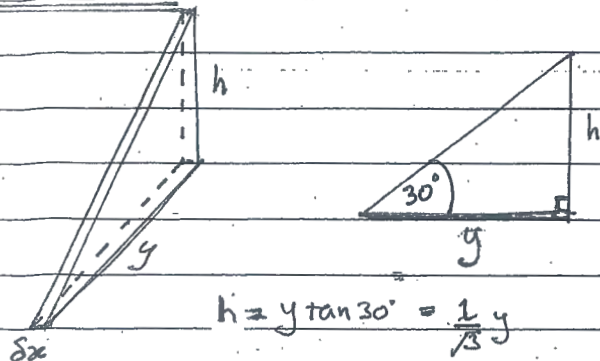
2 - marks - correct solution

1 - mark - one error

Asymptotes $y = \pm \frac{b}{a}x$ $m_{PR} = -\frac{a}{b}$ $m_{PQ} = \frac{a}{b}$ $\therefore$ equations are $bx - ay = 0$ $bx + ay = 0$ PR and PQ are perpendicular distances. $\text{dist} = \frac{ Ax_1 + By_1 + C }{\sqrt{a^2 + b^2}}$ $PR = \frac{ bx_1 + ay_1 }{\sqrt{a^2 + b^2}}$ $PQ = PR = \frac{ bx_1 - ay_1 }{\sqrt{a^2 + b^2}}$ $PR \cdot PQ = \frac{ bx_1 + ay_1 }{\sqrt{a^2 + b^2}} \times \frac{ bx_1 - ay_1 }{\sqrt{a^2 + b^2}}$ $= \frac{ b^2x_1^2 - a^2y_1^2 }{(a^2 + b^2)}$ $e^2 = 1 + \frac{b^2}{a^2} \Rightarrow a^2e^2 = a^2 + b^2$ $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$ $b^2x_1^2 - a^2y_1^2 = a^2b^2$ $PQ \cdot PR = \frac{a^2b^2}{a^2e^2} = \frac{b^2}{a^2}$	2 marks Denominator formed correctly as e^2 1 mark - correct expression for both distances
3 marks	
2 marks	
1 mark	

Question 16

a)



$$h = y \tan 30^\circ = \frac{1}{\sqrt{3}} y$$

$$A = \frac{1}{2} y h = \frac{1}{2\sqrt{3}} y^2$$

$$\delta V = A \delta x \Rightarrow \delta V = \frac{1}{2\sqrt{3}} y^2 \delta x$$

$$V = \int_{-5}^5 \frac{1}{2\sqrt{3}} y^2 dx = 2 \times \frac{1}{2\sqrt{3}} \int_0^5 y^2 dx$$

$$y^2 = 25 - x^2 \Rightarrow V = \frac{1}{\sqrt{3}} \int_0^5 (25 - x^2) dx$$

$$= \frac{1}{\sqrt{3}} \left[25x - \frac{x^3}{3} \right]_0^5$$

$$V = \frac{1}{\sqrt{3}} \left(125 - \frac{125}{3} \right) - 0$$

$$V = \frac{150}{3\sqrt{3}} = \frac{50\sqrt{3}}{9} \text{ u}^3$$

Marks

3 Marks

Correct solution

2 Marks

Obtains a

correct volume

element and

correct primitive

function

1 Mark

Obtains a

correct volume

element

Marks

1 Mark

Correct solution

with sufficient

working out

provided.

3 Marks

Correct solution

2 Marks

Attempts to

use

$$\int f(x) dx = \int f(x) dx$$

1 Mark

Attempts to use

integration by

parts and

obtains a correct

relevant

expression

b) (i)

$$x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$x=0 \Rightarrow \theta=0$$

$$x=1 \Rightarrow \theta = \pi/4$$

$$\int \frac{\log_e(1+x)}{1+x^2} dx = \int \frac{\log_e(1+\tan \theta) \sec^2 \theta d\theta}{1+\tan^2 \theta}$$

$$= \int \frac{\log_e(1+\tan \theta) \sec^2 \theta d\theta}{\sec^2 \theta}$$

$$= \int \log_e(1+\tan \theta) d\theta$$

$$(ii) I = \int_0^{\pi/4} \log_e(1+\tan \theta) d\theta = \int_0^{\pi/4} \log_e \{1+\tan(\frac{\pi}{4}-\theta)\} d\theta$$

$$1 + \tan(\frac{\pi}{4}-\theta) = \frac{1 + \tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \tan \theta}$$

$$= 1 + \frac{1 - \tan \theta}{1 + \tan \theta}$$

$$= \frac{1 + \tan \theta + 1 - \tan \theta}{1 + \tan \theta}$$

$$= \frac{2}{1 + \tan \theta}$$

$$I = \int_0^{\pi/4} \log_e \left\{ \frac{2}{1 + \tan \theta} \right\} d\theta$$

$$= \int_0^{\pi/4} \log_e 2 d\theta - \int_0^{\pi/4} \log_e(1 + \tan \theta) d\theta$$

$$I = \left[(\log_e 2) \theta \right]_0^{\pi/4} - I$$

$$2I = \frac{\pi \log_e 2}{4} - 0$$

$$I = \frac{\pi \log_e 2}{8}$$

$$\int_0^1 \frac{\log_e (1+x)}{1+x^2} dx = \frac{\pi \log_e 2}{8}$$

(c) (i) $\int \frac{x}{(1+x^2)^n} dx = \int x(1+x^2)^{-n} dx = \frac{1}{2} \int 2x(1+x^2)^{-n} dx$ 2 Marks

$$= \frac{1}{2} \left\{ \frac{(1+x^2)^{-n+1}}{-n+1} \right\} = \frac{1}{2} \left\{ \frac{(1+x^2)^{-(n-1)}}{-(n-1)} \right\}$$

$$= \frac{-1}{2(n-1)} \left\{ \frac{1}{(1+x^2)^{n-1}} \right\} + C$$

Correct solution

with sufficient working

1 Mark

Obtains a correct reverse chain rule integral expression.

(ii) $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx = \int_0^1 \frac{1+x^2-x^2}{(1+x^2)^n} dx$

$$= \int_0^1 \frac{1+x^2}{(1+x^2)^n} dx - \int_0^1 \frac{x^2}{(1+x^2)^n} dx$$

$$= \int_0^1 \frac{1}{(1+x^2)^{n-1}} dx - \int_0^1 x^2(1+x^2)^{-n} dx$$

3 Marks

Correct solution

2 Marks

Obtains expressions

$$I_n = I_{n-1} - \int_0^1 x^2(1+x^2)^{-n} dx \quad (1)$$

(1) and (2)

But $\int_0^1 x^2(1+x^2)^{-n} dx = \int_0^1 x \left\{ x(1+x^2)^{-n} \right\} dx$

1 Mark

Obtains expression

(1)

$$u = x \Rightarrow u' = 1$$

$$dv = x(1+x^2)^{-n} \Rightarrow v = \int x(1+x^2)^{-n} dx = \frac{1}{2(n-1)} \left\{ \frac{1}{1+x^2)^{n-1}} \right\}$$

Marks

$$\int_0^1 x^2(1+x^2)^{-n} dx = \left[\frac{-x}{2(n-1)(1+x^2)^{n-1}} \right]_0^1 - \int_0^1 \left\{ \frac{-1}{2(n-1)(1+x^2)^{n-1}} \right\} dx$$

Marks

$$= \frac{-1}{2(n-1)(2^{n-1})} - 0 + \frac{1}{2(n-1)} I_{n-1}$$

$$= \frac{-1}{2^n(n-1)} + \frac{1}{2(n-1)} I_{n-1} \quad (2)$$

$$I_n = I_{n-1} - \left\{ \frac{-1}{2^n(n-1)} + \frac{1}{2(n-1)} I_{n-1} \right\}$$

$$I_n = I_{n-1} + \frac{1}{2^n(n-1)} = \frac{1}{2(n-1)} I_{n-1}$$

$$= I_{n-1} - \frac{1}{2(n-1)} I_{n-1} + \frac{1}{2^n(n-1)}$$

$$= I_{n-1} \left\{ 1 - \frac{1}{2(n-1)} \right\} + \frac{1}{2^n(n-1)}$$

$$= \left\{ \frac{2n-2-1}{2(n-1)} \right\} I_{n-1} + \frac{1}{2^n(n-1)}$$

$$I_n = \left\{ \frac{2n-3}{2(n-1)} \right\} I_{n-1} + \frac{1}{2^n(n-1)}$$

(iii) $0 \leq x \leq 1 \Rightarrow (1+x^2) \geq 0$ and $(1+x^2) \leq (1+1) \leq 2$ 2 Marks

Correct solution

$$0 \leq x \leq 1 \Rightarrow 0 < (1+x^2)^n \leq 2^n$$

Marks

$$(1+x^2)^n \leq 2^n \Rightarrow \frac{1}{1+x^2} \geq \frac{1}{2^n}$$

1 Mark

$$\int_0^1 \frac{1}{1+x^2} dx \geq \int_0^1 \frac{1}{2^n} dx$$

Sufficient
relevant progress

$$\geq \frac{1}{2^n} [x]_0^1$$

using a valid
method.

$$\underline{\underline{I_n \geq \frac{1}{2^n} (1-0) \geq \frac{1}{2^n}}}$$